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Satellite Image Classification using CNN with Particle Swarm Optimization Classifier

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Abstract

Disaster relief, police work, and environmental monitoring all benefit from satellite images. Objects and infrastructure in the images must be manually identified for these applications. Due to the large areas that need to be searched and the limited number of accessible analysts, automation is essential. However, the accuracy and dependability of existing object recognition and classification algorithms renders them inadequate for the task. One family of machine learning algorithms called "deep learning" has showed immense potential for automating these kinds of jobs. Convolutional neural networks have been successful in the area of image recognition. Here, convolutional neural networks (CNNs) and a particle swarm optimization classifier is utilized to develop efficient algorithms for classifying satellite images. The results of this classifier model are better than those of existing approaches.

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*Keywords:*Deep Learning, Convolutional Neural Network, Particle Swarm Optimization, Satellite Image Classifier, Satellite Images

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1. Introduction

The rapid expansion of satellite-based remote sensing data has generated a growing interest in innovative techniques of data mining that can automate the process of extracting useful information from these massive databases. Different satellite operators and manufacturers use a wide range of remote sensing system specifications. Remote sensing data can be used in a variety of products and contexts. There are several potential uses for high-resolution satellite images. Agriculture, mapping, land use, tourism, environmental monitoring, and military are all examples of such uses. In addition, answers are found to a wide range of issues, like tracking the environment and the impact of human activity, finding polluted areas or illegal structures, gauging the health of a forest or plantation, tracking how much land is being used for development, and so on. In the remote sensing sector, as in many others, classification of the visual input is a crucial first step in solving any computer vision problem. There is a plethora of fresh research opportunities in remote sensing related to the classification of high-resolution satellite images. The VHR classification challenge has been greatly aided in recent years by the development of cutting-edge techniques. Any method of classification begins with the question of how to identify the feature being classified, followed by the question of how to use that identification to detect a new instance. Due to the impracticality of manual identification, significant effort has been invested over many years toward developing automatic and discriminatory visual feature descriptors. Machine learning approaches revolve around making comparisons between existing data and newly created objects. As land cover and land-use maps are required for multi-temporal study and offer valuable information to other processes, the remote sensing image classification process faces a number of challenges. The genera example image of satellite image classification is shown in the figure 1

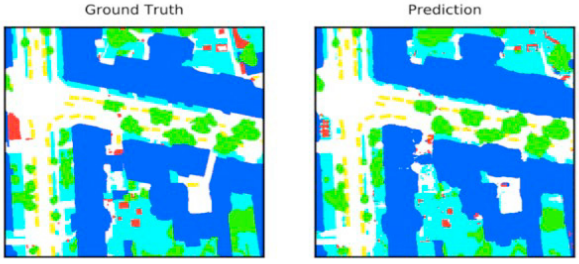


Figure: 1. General example image of Satellite image classification

Due to its improved feature representation, CNNs in particular have recently seen widespread adoption in a number of computer vision applications. This is due to CNNs' remarkable ability to automatically recognize interconnected contextual clues inherent in image classification challenges. The layers of a CNN are made up of trained convolution filters. In order to learn the hierarchical contextual picture features, these filters implement the conventional deep learning network structure. Deeply networked neural architecture was used to create CNN's specialized features. Thus, the features that are developed are symbolic representations of pixels in the original image. Therefore, feature selection has been delegated to the network infrastructure. Assembling the nonlinear processing units, then, is the job of all the different layers and neurons. Tree-like "layers" are used to teach adaptable feature representations. Lower-level features are trained in the first layer using the data as a guide, while higher-level features are trained in the subsequent layers. This allows the network to learn the multi-level features necessary for robust classifiers.

The past five years saw a proliferation of literature on employing CNNs for satellite imaging. Algorithms that employ CNNs have improved generalization capabilities due to a critical characteristic that eliminates the need for feature extraction in advance. CNNs have improved results in numerous scenarios. Recent research has demonstrated that CNNs can accurately identify and classify images. The remote sensing community has also

benefited greatly from deep learning's application to specialized data types. In this article, a CNN model with PSO classifier is introduced for satellite image classification.

- ✓ The remainder of the paper is organized as follows:
- ✓ Section 2 introduces existing techniques, reviews their benefits, and limits, and summarizes the related works that targeted the same problem in literature.
- ✓ Section 3 describes the satellite image classification approach using PSO in the current study.
- ✓ Section 4 is reserved for the discussion of the experimental results.
- ✓ Finally, our work is concluded with a description of the possible extensions in Section 5

2. Literature Survey

Numerous neural network related methods necessitate training with thousands of images, which poses a challenge in terms of available training sets. Model performance is also quite sensitive to changes in the computing environment, frameworks, hardware, and compilers. Hence new algorithms are explored to have accurate classification

The utilization of a flexible Convolutional Neural Network (CNN) architecture, known as Sat CNN, has been suggested as a valuable approach for scene categorization of High Spatial Resolution Remote Sensing (HSR-RS) images [1]. The foundation of this efficient convolutional neural network (CNN) design lies in the innovative research conducted in the field. This research has led us to utilize convolutional layers with smaller kernels, resulting in enhanced efficiency. The conducted experiments on SAT data sets have shown that SatCNN is capable of efficiently and rapidly extracting robust features to handle the intra-class variability, even when using small convolutional kernels. Additionally, the deeper convolutional layers enable the spontaneous modeling of the relative spatial relationships.[2] uses an ensemble classifier to identify bodies of water in satellite pictures for use in gauging the severity of floods. Using both the bands themselves and the Normalized Difference Water Index (NDWI), a classification was achieved. It is clear from the results that selecting suitable classifiers to ensemble is crucial for improving classification accuracy using ensemble classifiers. Furthermore, the results demonstrate that this method achieves high accuracy in both visual and non-visual location categorization when using bands and NDWI, respectively.

To classify high-resolution satellite images, a framework built on top of an enhanced architecture that combines a conventional neural network structure with custom-made features [3] demonstrated that by adding in some carefully built features, a CNN can outperform much bigger, unaugmented CNN architectures when it comes to discriminating between satellite images. For the SAT-4 and SAT-6 datasets, this framework achieves higher classification accuracy than any other known methods. The paper [4] introduces an effective approach to address privacy concerns associated with satellite image data. It proposes the utilization of privacy-preserving deep learning (PPDL) approaches in conjunction with publicly accessible deep learning (DL) models. This study proposes the utilization of a partially homomorphic encryption system, specifically the Paillier scheme, as a means to enable the secure processing of sensitive data while minimizing the potential for unauthorized exposure. When applied to a specialized CNN or to preexisting transfer learning techniques, this method produces reliable results. Because of the low computational overhead of the proposed encryption approach, CNN models can be trained directly on encrypted data.

With only a fraction of the computational resources required, [5] analyzes the effectiveness of the SimpleCNN architecture on the SAT-4 and SAT-6 aerial dataset for land cover categorization. It is the primary step while the combined Red and NIR band was used in the second test scenario, the NIR band was used in the SAT-6 (Red and NIR) confusion matrix due to its greater suitability for sensing vegetation. The proposed model functioned better

when both frequencies were considered simultaneously. Over 96% accuracy in classification was achieved, which is a significant improvement.

Each pixel in the spectral domain is processed through a framework based on a 1D convolution neural network to extract descriptive local features for improved classification, as described in [6-8]. Experimental results demonstrate the superior classification accuracy of the proposed method when compared to both traditional unmixing-based and neural network approaches, even when employing a smaller sample size.[12] devised a novel approach that combines differential privacy (DP) and secure multi-party computation (SMPC). The researchers showcased that when the number of parties is increased and the data sections are decreased, the utilization of differential privacy (DP) results in diminished accuracy. Additionally, they demonstrated that the utilization of SPMC introduces potential vulnerabilities during the inference stage. In order to address these challenges, the researchers devised an improved federated learning (FL) framework that integrates differential privacy (DP) and secure multi-party computation (SMPC). The system that has been implemented exhibits scalability, robustness against potential threats from adversaries, and effectively maintains a high level of accuracy in the model. The authors employed a federated learning approach to train the model, ensuring privacy preservation while maintaining accuracy. In a separate investigation, [13-20] developed a confidential collaborative framework for machine learning by integrating secure multiparty computation (SMPC) and differential privacy (DP). The utilization of differential privacy (DP) is employed to ensure privacy protection, whereas the machine learning model was developed based on neural networks.

3. Proposed Work

In this section, the procedure of preparing images is initiated from both datasets for utilization as training data by extracting feature vectors through the application of a CNN. This model utilizes two unique datasets: the SAT airborne colour picture datasets, which consist of 28*28 uint8 type images, and the UCMD dataset, which consists of 256*256 uint8 type images with distinct bands for red, green, and blue. In order to develop a classification model, it is important to initially convert all satellite images to greyscale. Subsequently, the features vector corresponding to each image in the training set must be extracted. The reduction of the near-infrared (NIR) band in the SAT datasets is undertaken. CNNs are employed for the purpose of feature extraction. The performance of a satellite image classification system is contingent upon the robustness of the features extracted from the training dataset. The testing phase is the stage at which the genuine efficacy of these features will become evident.

In this study, a conventional approach is provided for extracting features from images, which provides a comprehensive dataset for the CNN to acquire knowledge from. Each layer in the CNN generates an activation or reaction when presented with an input image. It is probable that only a limited number of layers inside the CNN architecture are suitable for extracting characteristics from the input image. The features that have been extracted from the lower layers can be utilized as training features due to their ability to provide more advanced features. On the contrary, the initial layer of the CNN is responsible for capturing fundamental visual characteristics such as edges and blobs in the input image. Within the framework of CNN, the initial layer assumes the crucial role of acquiring the ability to identify and discern edge and blob characteristics. These acquired features are subsequently subjected to further processing by subsequent layers that possess more depth. In this scenario, the primary characteristics are integrated with comprehensive high-level features within the fully connected layer, enabling their utilization in tasks such as recognition or classification. The workflow of the proposed model is explained in the fig 2,

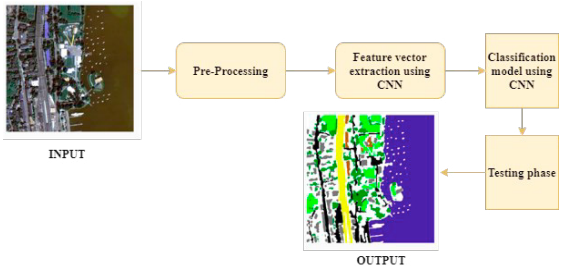


Figure 2: Workflow of proposed model

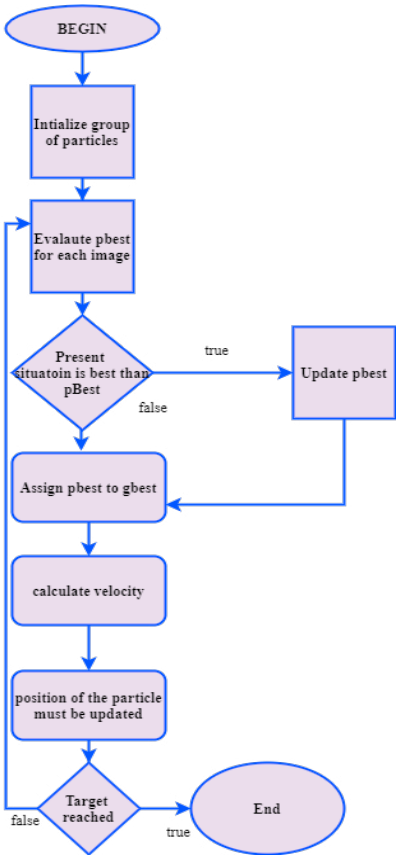


Fig 3:Overview of PSO

3.2 PSO

The following are the primary procedures for image classification using the proposed approach that shown in the fig 3.

a. Starting with an input image of type, randomly place artificial swarms on it. The search parameters and their associated others are being initialized.
b. Using only the force of gravity from outside the system, particle is able to identify the pixel in which it is located. Using the information from the searching model, the particle moves on to the next pixel. All gravitations for the pixels whose space the particle has traversed are updated as part of the search procedure.

c. The gravitation of congeneric pixels is larger, so the gravitation of following massive particles continuously improves the gravitation of congeneric pixels, allowing for the classification of congeneric pixels and non-congener pixels through the measurement of gravitation.
d. Particle will embark on a stochastic search for a new path if its excitement value exceeds the threshold value. This will aid in global searching and prevent the process from stumbling into the territory of local searching.

Particle swarms of every flavour bring their own brand of self-organization and cooperation to the table throughout the sorting process. Each particle is assigned a random initial velocity and position at the beginning. One of the most distinctive aspects of the PSO method is its velocity, which is the driving force behind the evolution of a particle's position in the search for optimal solutions. A popular method for controlling the speed is to multiply the particle's velocity by a factor. To strike a compromise between the two extremes, this speed regulation is in place. The following formula is used to calculate the velocity.

$$v_{T+1} = v_T + r_1 c_1 (p - x) + r_2 c_2 (g - x)$$

Every particle must preserve its position. It is referred as p_{best} (local best position) and the g_{best} (global best position) among all particles. The pbest is assigned using best position of neighbourhood. Thereafter be a phase of evaluating the satellite image categorization model is implemented. The remaining 30% of each dataset is used to evaluate the classifier's performance here. The testing images will undergo the same steps in preparing the input data for training, including pre-processing, and extracting a set of features for all categories in the datasets, and saving the data as two-dimensional matrices where each row corresponds to a single image.

4. Results and Discussion

4.1 Dataset –SAT6

The dataset provided in this iteration has a total of 405,000 images, with each image having dimensions of 28 × 28 pixels. These images encompass six distinct geographical classifications, namely barren ground, trees, grassland, roads, buildings, and water bodies. A total of 324,000 images were selected as the training dataset, while the remaining 81,000 images were designated for the testing dataset.

4.2 Performance analysis

The SAT-6 dataset is used to evaluate the suggested methodology. The same data set is used to report the outcomes of alternative popular architectures. Classifying satellite image data is an effortless with PSO classifier. Among all reported architectures, PSO classifier offers the highest accuracy. However, in circumstances when the classes are unbalanced or there are several classes, accuracy alone may not be the optimal statistic for evaluating the effectiveness of a machine learning model shown in table 1.

Table 1 Comparative analysis of proposed work (Performance Metrics)

	ACCURACY	PRECISOIN	RECALL	F1-SCORE
CNN [9]	0.89	0.84	0.74	0.83
DCNN [10]	0.93	0.92	0.82	0.96
PPDL [11]	0.95	0.96	0.89	0.98
Proposed	0.99	0.97	0.93	0.99

❖ **Accuracy**

This metric measures the proportion of correctly classified information instances relative to the total number of data instance.

$$Accuracy = \frac{TN + FP}{TP + FP + TN + FN}$$

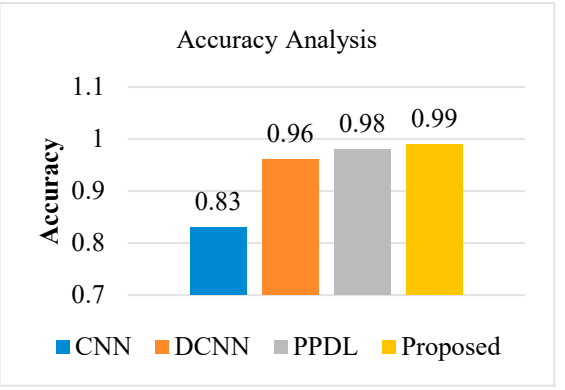


Figure 4: Accuracy

This classifier model outperforms the best in terms of accuracy when compared to existing approached as show in figure 4.

❖ **Precision**

It is defined as the proportion of correct predictions to all model predictions.

$$Precision = \frac{TP}{TP + FP}$$

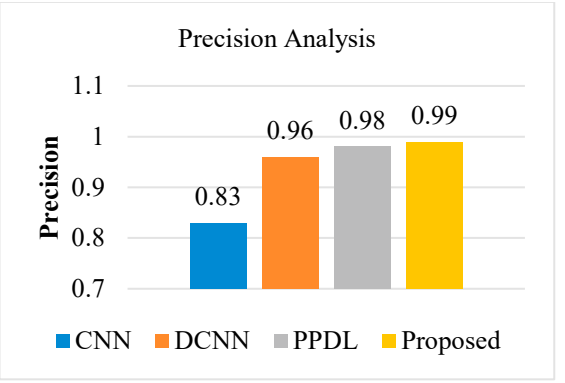


Figure 5: Precision

When compared to state-of-the-art classifier models, this one achieves the highest precision value is shown in figure 5

❖ **Recall**

Recall is determined by dividing the number of positive results by the sum of the positive and negative results.

$$Recall = \frac{truepositive}{truepositive + falsenegative}$$

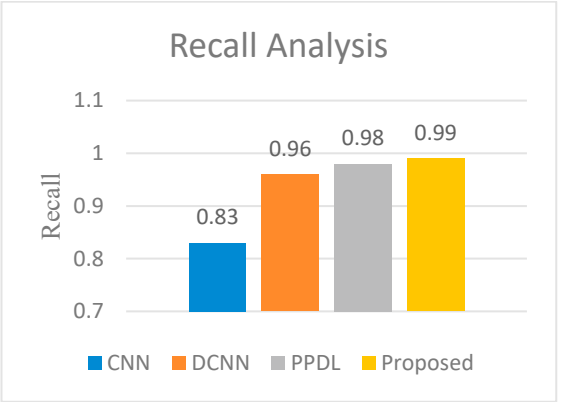


Figure 6: Recall

As can be seen in figure. 6, this proposed method achieves better recall value than any of the other methods currently in use.

❖ **F1-score**

Precision and recall are weighted equally in the F1 score. It involves consideration of both false positives and false negatives since these can occur in both accuracy and recall.

$$F1 - score = \frac{2 * (Recall * precsion}{(Recall + Precision)}$$

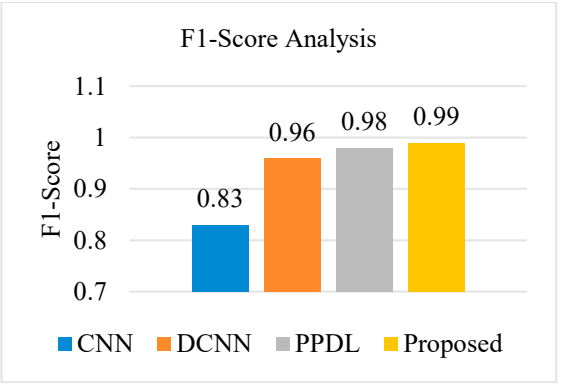


Figure 7: F1-score

The classifier model has superior performance in terms of F1-score when compared to existing approaches, as illustrated in figure 7.

4.3 Discussion

Overfitting, computational complexity, and vanishing or ballooning gradients are just a few of the problems that deeper neural networks are able to perform into. When approached with limited or noisy data, very deep networks may struggle to acquire meaningful representations and demand large computational resources for training. Overfitting is a problem in machine learning, and deeper networks may be more vulnerable to it when the dataset is limited, or the model is complicated. Therefore, while developing neural networks for specific tasks and datasets, it is essential to carefully assess the trade-offs between depth and performance. PSO Classifier is better at handling cases of vanishing gradients. This is because it improves the convergence qualities of complex neural networks and allows for the training of deeper networks.

5. Conclusion

The use of satellite images is crucial in many fields, such as emergency management, policing, and environmental observation. For these uses, it is necessary to manually recognize specific features and structures in images. Automation is necessary since there are large areas to search throughout and a limited number of analysts to execute the searches. However, the accuracy and dependability of existing algorithms for object recognition and classification offer scope to explore new techniques. There is hope that these kinds of jobs can be automated with the help of the deep learning family of machine learning algorithms. Using convolutional neural networks, it has been able to successfully interpret images. Here, efficient techniques have been developed for classifying satellite images using a combination of convolutional neural networks and a particle swarm optimization classifier. This classifier model provides better results.

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